# HUICHON, Korea, Democratic People's Republic of

WMO: **470390**

Lat: **40.167N** Lon: **126.250E** Elev: **155** StdP: **99.48** Time Zone: **9.00 (E09)** Period: **90-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-19.5** | **-17.2** | **-22.7** | **0.5** | **-17.6** | **-20.5** | **0.6** | **-15.9** | **5.5** | **-7.1** | **4.5** | **-7.6** | **0.2** | **320** | **0.392** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **7.7** | **31.7** | **25.1** | **30.4** | **24.3** | **29.1** | **23.5** | **26.7** | **30.5** | **25.6** | **29.0** | **24.6** | **27.7** | **1.4** | **230** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **25.6** | **21.2** | **29.6** | **24.5** | **19.9** | **28.1** | **23.7** | **18.8** | **26.9** | **85.0** | **30.4** | **79.9** | **29.0** | **75.9** | **27.7** | **31.3** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **5.3** | **4.3** | **3.4** | DB | **-22.5** | **34.5** | **3.0** | **1.2** | **-24.7** | **35.3** | **-26.4** | **36.0** | **-28.1** | **36.7** | **-30.2** | **37.5** | (4) |
| (5) | | | | WB | **-22.6** | **28.3** | **2.9** | **1.5** | **-24.7** | **29.3** | **-26.3** | **30.2** | **-27.9** | **31.0** | **-30.0** | **32.1** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | DBAvg | | **9.4** | **-8.1** | **-4.2** | **2.5** | **9.5** | **16.0** | **21.0** | **24.2** | **24.2** | **18.8** | **11.3** | **2.5** | **-5.6** | (6) |
| (7) | | DBStd | | **11.83** | **4.34** | **4.18** | **3.80** | **3.83** | **2.98** | **2.40** | **2.12** | **2.36** | **3.06** | **3.71** | **4.87** | **4.52** | (7) |
| (8) | | HDD10.0 | | **1988** | **561** | **398** | **235** | **54** | **1** | **0** | **0** | **0** | **0** | **29** | **227** | **482** | (8) |
| (9) | | HDD18.3 | | **3760** | **819** | **631** | **492** | **265** | **83** | **6** | **0** | **1** | **31** | **217** | **474** | **741** | (9) |
| (10) | | CDD10.0 | | **1779** | **0** | **0** | **1** | **39** | **188** | **331** | **441** | **441** | **264** | **71** | **3** | **0** | (10) |
| (11) | | CDD18.3 | | **510** | **0** | **0** | **0** | **1** | **12** | **87** | **182** | **183** | **46** | **1** | **0** | **0** | (11) |
| (12) | | CDH23.3 | | **3729** | **0** | **0** | **0** | **14** | **180** | **622** | **1270** | **1351** | **283** | **9** | **0** | **0** | (12) |
| (13) | | CDH26.7 | | **1061** | **0** | **0** | **0** | **1** | **32** | **158** | **388** | **439** | **41** | **1** | **0** | **0** | (13) |
| (14) | Wind | WSAvg | | **0.7** | **0.5** | **0.8** | **0.9** | **1.1** | **1.1** | **0.8** | **0.7** | **0.6** | **0.5** | **0.6** | **0.6** | **0.5** | (14) |
| (15) | Precipitation | PrecAvg | | **1141** | **9** | **20** | **27** | **55** | **84** | **122** | **338** | **254** | **109** | **54** | **49** | **22** | (15) |
| (16) | | PrecMax | | **1963** | **32** | **96** | **89** | **97** | **156** | **268** | **1016** | **1065** | **464** | **129** | **123** | **59** | (16) |
| (17) | | PrecMin | | **662** | **0** | **2** | **1** | **5** | **17** | **43** | **95** | **52** | **13** | **3** | **9** | **3** | (17) |
| (18) | | PrecStd | | **336** | **8** | **24** | **19** | **25** | **38** | **57** | **190** | **193** | **92** | **30** | **30** | **14** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **4.1** | **9.5** | **17.2** | **25.4** | **29.8** | **32.2** | **34.1** | **33.8** | **29.5** | **24.5** | **16.6** | **5.9** | (19) |
| (20) | | | MCWB | | **1.3** | **5.2** | **10.2** | **16.1** | **19.9** | **22.0** | **27.3** | **26.9** | **22.6** | **17.2** | **11.9** | **3.3** | (20) |
| (21) | | 2% | DB | | **1.7** | **6.4** | **14.0** | **22.4** | **27.2** | **30.1** | **31.7** | **31.9** | **27.9** | **21.9** | **13.6** | **4.0** | (21) |
| (22) | | | MCWB | | **-0.3** | **2.9** | **8.3** | **14.4** | **18.6** | **21.7** | **25.6** | **25.9** | **21.3** | **15.8** | **10.1** | **2.4** | (22) |
| (23) | | 5% | DB | | **0.3** | **4.4** | **11.4** | **20.0** | **25.3** | **28.5** | **30.3** | **30.5** | **26.4** | **20.0** | **11.6** | **2.3** | (23) |
| (24) | | | MCWB | | **-1.5** | **1.6** | **6.8** | **12.7** | **17.5** | **21.3** | **25.1** | **24.7** | **20.3** | **14.5** | **8.9** | **0.7** | (24) |
| (25) | | 10% | DB | | **-1.3** | **2.6** | **9.2** | **17.6** | **23.3** | **26.9** | **28.8** | **29.2** | **25.0** | **18.0** | **9.6** | **0.9** | (25) |
| (26) | | | MCWB | | **-2.9** | **0.2** | **5.4** | **11.4** | **16.4** | **20.7** | **24.4** | **24.0** | **19.5** | **13.2** | **7.4** | **-0.3** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **1.9** | **5.9** | **11.4** | **17.1** | **21.5** | **24.5** | **29.0** | **29.2** | **24.5** | **18.4** | **12.5** | **4.6** | (27) |
| (28) | | | MCDB | | **3.9** | **9.2** | **16.1** | **23.5** | **27.3** | **28.7** | **32.7** | **32.4** | **27.8** | **22.9** | **15.3** | **5.2** | (28) |
| (29) | | 2% | WB | | **0.0** | **3.3** | **8.9** | **15.0** | **19.4** | **23.2** | **26.9** | **26.9** | **22.7** | **16.5** | **10.7** | **2.4** | (29) |
| (30) | | | MCDB | | **1.4** | **5.7** | **12.9** | **21.3** | **25.4** | **28.1** | **30.6** | **30.8** | **26.0** | **20.5** | **13.0** | **3.7** | (30) |
| (31) | | 5% | WB | | **-1.3** | **1.8** | **7.2** | **13.4** | **18.2** | **22.4** | **25.7** | **25.8** | **21.5** | **15.3** | **9.1** | **1.0** | (31) |
| (32) | | | MCDB | | **0.1** | **4.1** | **11.1** | **19.1** | **24.1** | **27.0** | **29.1** | **29.3** | **25.1** | **18.8** | **11.4** | **2.1** | (32) |
| (33) | | 10% | WB | | **-2.8** | **0.4** | **5.7** | **11.7** | **17.1** | **21.4** | **24.8** | **24.8** | **20.3** | **13.9** | **7.5** | **-0.2** | (33) |
| (34) | | | MCDB | | **-1.3** | **2.3** | **8.9** | **16.5** | **22.3** | **25.6** | **27.8** | **27.9** | **23.8** | **17.2** | **9.3** | **0.9** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **11.5** | **11.1** | **10.5** | **12.2** | **12.1** | **10.4** | **7.7** | **9.0** | **11.3** | **12.5** | **9.4** | **9.6** | (35) |
| (36) | | 5% DB | MCDBR | | **11.5** | **12.1** | **14.4** | **16.7** | **16.2** | **13.7** | **10.6** | **11.0** | **12.9** | **15.0** | **11.9** | **9.2** | (36) |
| (37) | | | MCWBR | | **9.4** | **8.8** | **9.5** | **9.4** | **8.6** | **7.0** | **6.2** | **6.1** | **7.4** | **9.0** | **8.5** | **7.4** | (37) |
| (38) | | 5% WB | MCDBR | | **10.5** | **10.9** | **13.4** | **15.2** | **14.3** | **11.0** | **8.7** | **8.9** | **10.8** | **13.0** | **10.8** | **8.3** | (38) |
| (39) | | | MCWBR | | **8.9** | **8.2** | **9.3** | **9.4** | **8.4** | **6.7** | **5.9** | **5.8** | **6.9** | **8.4** | **8.1** | **7.0** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.379** | **0.456** | **0.527** | **0.602** | **0.627** | **0.688** | **0.645** | **0.556** | **0.486** | **0.510** | **0.452** | **0.391** | (40) |
| (41) | | taud | | | **1.982** | **1.840** | **1.740** | **1.656** | **1.649** | **1.599** | **1.742** | **1.945** | **2.051** | **1.876** | **1.933** | **1.993** | (41) |
| (42) | | Ebn at Noon | | | **731** | **721** | **715** | **695** | **691** | **650** | **675** | **726** | **751** | **661** | **642** | **678** | (42) |
| (43) | | Edh at Noon | | | **126** | **169** | **208** | **242** | **248** | **262** | **225** | **179** | **151** | **161** | **133** | **116** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **2.53** | **3.37** | **4.35** | **4.88** | **5.50** | **5.10** | **4.26** | **4.42** | **4.33** | **3.36** | **2.23** | **2.06** | (44) |
| (45) | | RadStd | | | **0.13** | **0.23** | **0.33** | **0.39** | **0.47** | **0.52** | **0.55** | **0.51** | **0.39** | **0.24** | **0.26** | **0.14** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (20 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page